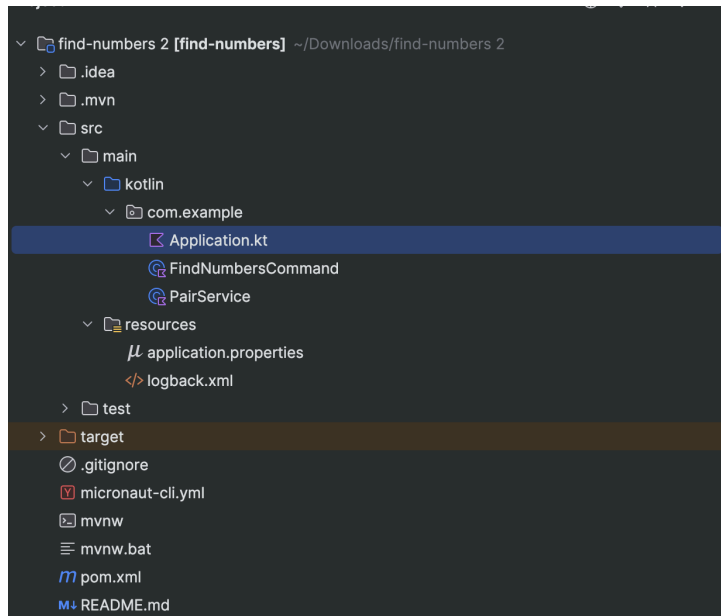


Pornind de la aplicația exemplu pentru ciurul lui Eratostene din laboratorul 11, să se modifice aceasta astfel încât să se găsească (dacă există) perechile de numere care satisfac $a * b = a + b * 3$ (această verificare va fi făcută în funcție) unde a aparține lui A , b aparține lui B , iar A și B sunt două ADT-uri cu câte 100 de valori inițializate aleator (restul în programul principal). Dacă se consideră necesar se mai pot crea oricâte servicii se dorește conform gândirii studentului.



NOTITE

1. Mergi pe site-ul asta: <https://micronaut.io/launch> si generezi proiectul dupa setarile astea

Application Type Command Line Application	Java Version 21	Name find-numbers	Base Package com.example
Micronaut Version ☒ 4.8.3 ☐ 4.8.4-SNAPSHOT ☐ 3.10.1	Language ☐ Java ☐ Groovy ☒ Kotlin	Build Tool ☐ Gradle ☐ Gradle Kotlin ☒ Maven	Test Framework ☒ JUnit ☐ Spock ☐ Kotest
+ FEATURES	+ DIFF	Q PREVIEW	🚀 GENERATE PROJECT

2. Dezarchiveaza
3. Modifica Pom.xml-ul dupa cum vezi mai jos

Pom.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/xsd/maven-4.0.0.xsd">

  <modelVersion>4.0.0</modelVersion>
  <groupId>com.example</groupId>
```

```
<artifactId>find-numbers</artifactId>
<version>0.1</version>
<packaging>jar</packaging>

<parent>
  <groupId>io.micronaut.platform</groupId>
  <artifactId>micronaut-parent</artifactId>
  <version>4.8.3</version>
</parent>

<properties>
  <jdk.version>17</jdk.version>
  <release.version>21</release.version>
  <kotlin.version>1.9.25</kotlin.version>
  <micronaut.version>4.8.3</micronaut.version>
  <micronaut.core.version>4.8.3</micronaut.core.version>
  <picocli.version>4.7.5</picocli.version>
  <micronaut.serialization.version>2.14.1</micronaut.serialization.version>
  <exec.mainClass>com.example.ApplicationKt</exec.mainClass>
</properties>

<dependencies>
  <dependency>
    <groupId>info.picocli</groupId>
    <artifactId>picocli</artifactId>
    <version>${picocli.version}</version>
    <scope>compile</scope>
  </dependency>
  <dependency>
    <groupId>io.micronaut</groupId>
    <artifactId>micronaut-context</artifactId>
  </dependency>
  <dependency>
    <groupId>io.micronaut.kotlin</groupId>
    <artifactId>micronaut-kotlin-runtime</artifactId>
  </dependency>

  <dependency>
    <groupId>io.micronaut.picocli</groupId>
    <artifactId>micronaut-picocli</artifactId>
  </dependency>

  <dependency>
    <groupId>io.micronaut.serde</groupId>
    <artifactId>micronaut-serde-jackson</artifactId>
  </dependency>

  <dependency>
    <groupId>org.jetbrains.kotlin</groupId>
```

```
<artifactId>kotlin-stdlib-jdk8</artifactId>
</dependency>

<dependency>
  <groupId>org.jetbrains.kotlin</groupId>
  <artifactId>kotlin-reflect</artifactId>
</dependency>

<dependency>
  <groupId>ch.qos.logback</groupId>
  <artifactId>logback-classic</artifactId>
  <scope>runtime</scope>
</dependency>

<dependency>
  <groupId>io.micronaut.test</groupId>
  <artifactId>micronaut-test-junit5</artifactId>
  <scope>test</scope>
</dependency>

<dependency>
  <groupId>org.junit.jupiter</groupId>
  <artifactId>junit-jupiter-engine</artifactId>
  <scope>test</scope>
</dependency>
</dependencies>

<build>
  <sourceDirectory>src/main/kotlin</sourceDirectory>
  <testSourceDirectory>src/test/kotlin</testSourceDirectory>

  <plugins>
    <plugin>
      <groupId>org.jetbrains.kotlin</groupId>
      <artifactId>kotlin-maven-plugin</artifactId>
      <version>${kotlin.version}</version>
      <executions>
        <execution>
          <id>compile</id>
          <goals>
            <goal>compile</goal>
            <goal>test-compile</goal>
          </goals>
        </execution>
      </executions>
      <configuration>
        <jvmTarget>${jdk.version}</jvmTarget>
        <compilerPlugins>
          <plugin>all-open</plugin>
        </compilerPlugins>
      </configuration>
    </plugin>
  </plugins>
</build>
```

```

        </compilerPlugins>
        <pluginOptions>
            <option>all-open:annotation=io.micronaut.aop.Around</option>
        </pluginOptions>
    </configuration>
    <dependencies>
        <dependency>
            <groupId>org.jetbrains.kotlin</groupId>
            <artifactId>kotlin-maven-allopen</artifactId>
            <version>${kotlin.version}</version>
        </dependency>
    </dependencies>
</plugin>

<plugin>
    <groupId>io.micronaut.maven</groupId>
    <artifactId>micronaut-maven-plugin</artifactId>
</plugin>

<plugin>
    <groupId>org.apache.maven.plugins</groupId>
    <artifactId>maven-compiler-plugin</artifactId>
    <configuration>
        <source>${jdk.version}</source>
        <target>${jdk.version}</target>
        <proc>none</proc>
    </configuration>
</plugin>
</plugins>
</build>
</project>

```

[Application.kt](#)

```

package com.example

fun main(args: Array<String>) {
    val pairService = PairService()
    val command = FindNumbersCommand(pairService)
    command.run()
}

```

[FindNumbersCommand.kt](#)

```

package com.example

```

```

import picocli.CommandLine.Command

@Command(
    name = "find-numbers",
    description = ["Find pairs where  $a * b = a + 3 * b$ "],
    mixinStandardHelpOptions = true
)
class FindNumbersCommand(
    private val pairService: PairService
) : Runnable {

    override fun run() {
        pairService.findPairs()
    }
}

```

[PairService.kt](#)

```

package com.example

import kotlin.random.Random

class PairService {
    fun findPairs() {
        val A = MutableList(100) { Random.nextInt(1, 1000) }
        val B = MutableList(100) { Random.nextInt(1, 1000) }

        A[0] = 6
        B[0] = 2
        A[1] = 4
        B[1] = 4

        println("Perechi care satisfac relația:  $a * b = a + 3 * b$ ")
        println()

        var found = false
        for (a in A) {
            for (b in B) {
                if (a * b == a + 3 * b) {
                    found = true
                    println("a = $a, b = $b")
                }
            }
        }
        if (!found) {
            println("Nu s-a găsit nicio pereche care să satisfacă relația.")
        }
    }
}

```

```
}  
}
```